

RainRisk: Predicting Meteorological Drought Risk Across India from 117 Years of Rainfall Records



An end-to-end, leak-free machine learning pipeline classifying drought risk across 36 Indian meteorological subdivisions, built on historical monsoon rainfall and planetary ocean–atmosphere teleconnections.

By Rajratna Bipin Dhiwar

Erratic monsoon rainfall threatens agriculture, water security, and rural livelihoods across India, where no accessible, transparent ML-based drought classification exists at the academic level.

56.0%

EXACT ACCURACY,
RANDOM FOREST

93.0%

Off-by-one Accuracy

61 / 61

Automated tests passing
zero leakage, chronological split

WHY THIS MATTERS

The Problem & The Data

Erratic monsoon rainfall threatens agriculture, water security, and rural livelihoods across India. IMD has published over a century of rainfall records, but no accessible, transparent, ML-based drought risk classification exists at the academic or operational level.

RainRisk builds a reproducible pipeline that classifies Indian subdivisions and years into IMD's own operational drought risk categories, using classical, interpretable machine learning.

117 years

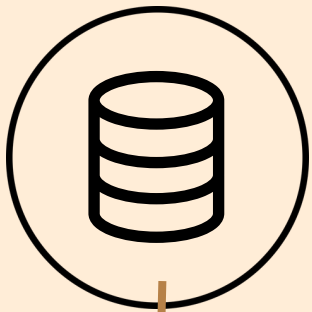
of IMD rainfall records (1901 to 2017), across 36 meteorological subdivisions, totaling 3,913 subdivision-years.

2 ocean indices

Cross-referenced with NOAA Niño 3.4 SST and JAMSTEC Indian Ocean Dipole teleconnection indices, planetary signals that precede the monsoon.

HOW IT WORKS

Method



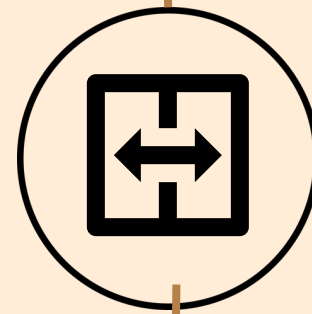
Raw IMD rainfall data
Monthly + seasonal totals, 1901–2017, 36 subdivisions



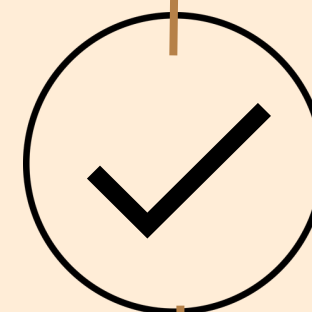
IMD 6-category labeling
No Rainfall, Large Deficient, Deficient, Normal, Excess, Large Excess: departure bands confirmed against IMD's own bulletin



23-feature engineering
Monthly & seasonal lags, rolling 3/5-yr windows, monsoon concentration ratios, plus 5 zero-leakage ENSO / IOD teleconnection features



Chronological train / test split
Train on 1901 to 2010, test on 2011 to 2017 (243 held-out observations), never shuffled



SMOTENC class balancing
Fit on the train fold only; synthetic rows always resolve to a real subdivision, never a fractional blend



Random Forest classifier
200 trees, max depth 12, min-leaf 2, the best of 5 benchmarked architectures

5 models benchmarked on the identical 23-feature set: Logistic Regression, Random Forest, SVM, Gradient Boosting, and HistGradientBoosting, each evaluated with the same chronological split and the same balanced-accuracy and macro-F1 metrics.

WHAT WE FOUND

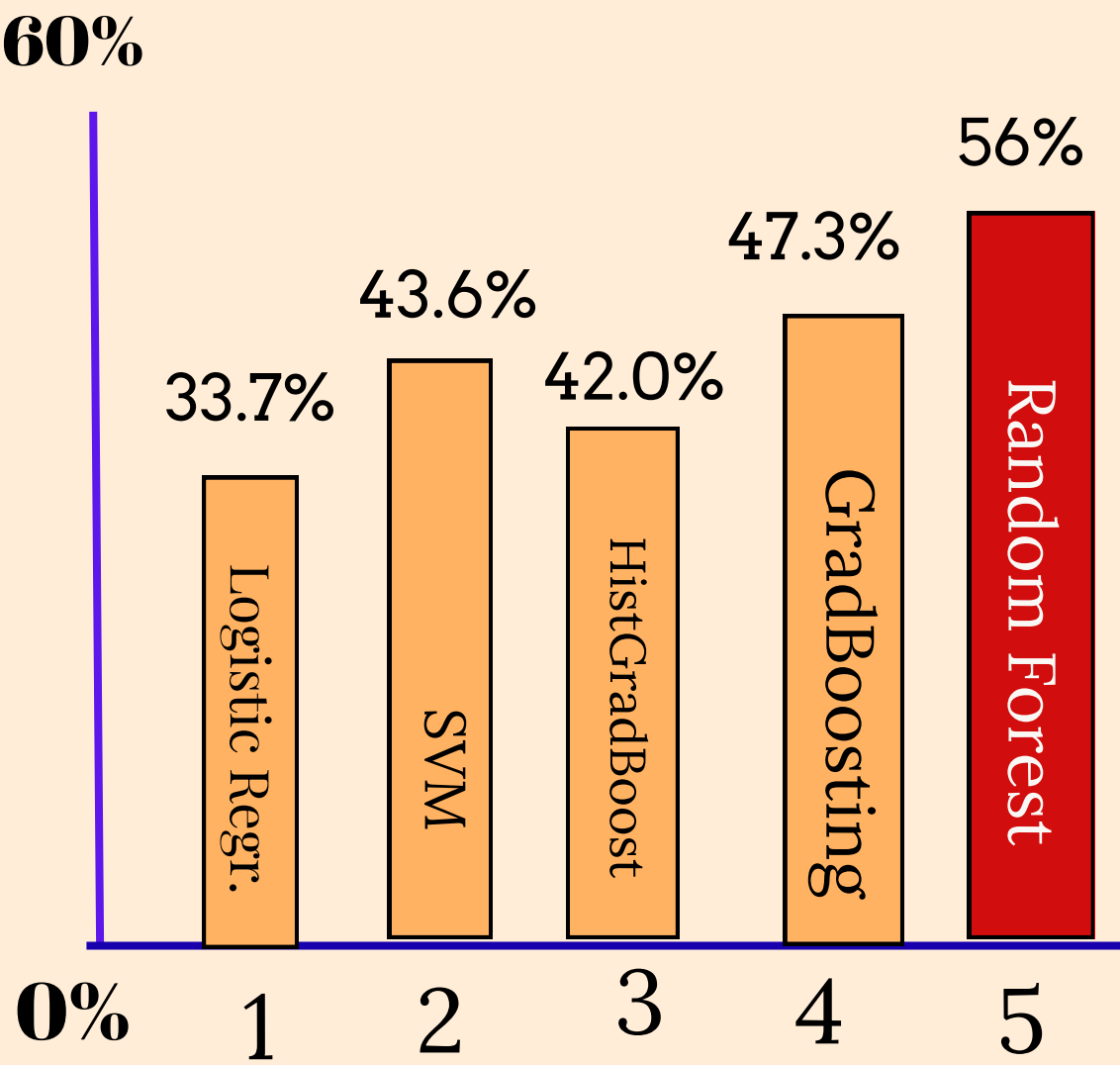
Results

56.0%/93.0%

Exact accuracy and off-by-one accuracy for Random Forest on held-out 2011 to 2017 test data (N=243), against 33.7% exact accuracy for the baseline Logistic Regression.

Model comparison

Exact accuracy on held-out test set, 23-feature teleconnection architecture



Scan the QR for
Repository Link

Platform, Rigor & Takeaway

Interactive web platform

A decoupled FastAPI + React 18 single-page application with a live, interactive map of India, 117-year timeline exploration, and a real-time what-if teleconnection scenario simulator.

Rigorously tested

61 / 61 automated tests passing across 7 modules. Every design decision, from thresholds to split strategy to model choice, is traceable to either IMD's own operational standard or a cited source from a 25-paper literature review.

Takeaway

RainRisk predicts regional Indian monsoon drought months in advance by combining 117 years of historical rainfall with Pacific and Indian Ocean temperatures (ENSO & IOD). Using a leakage-free Random Forest model, it achieves 93% off-by-one accuracy and automatically generates actionable crop advisories for farmers.

A transparent, reproducible template for climate-risk ML, where honesty about a model's limitations is treated as a result, not a flaw to hide.